Student:	Email:
Kyah Owens	ndowen02@louisville.edu

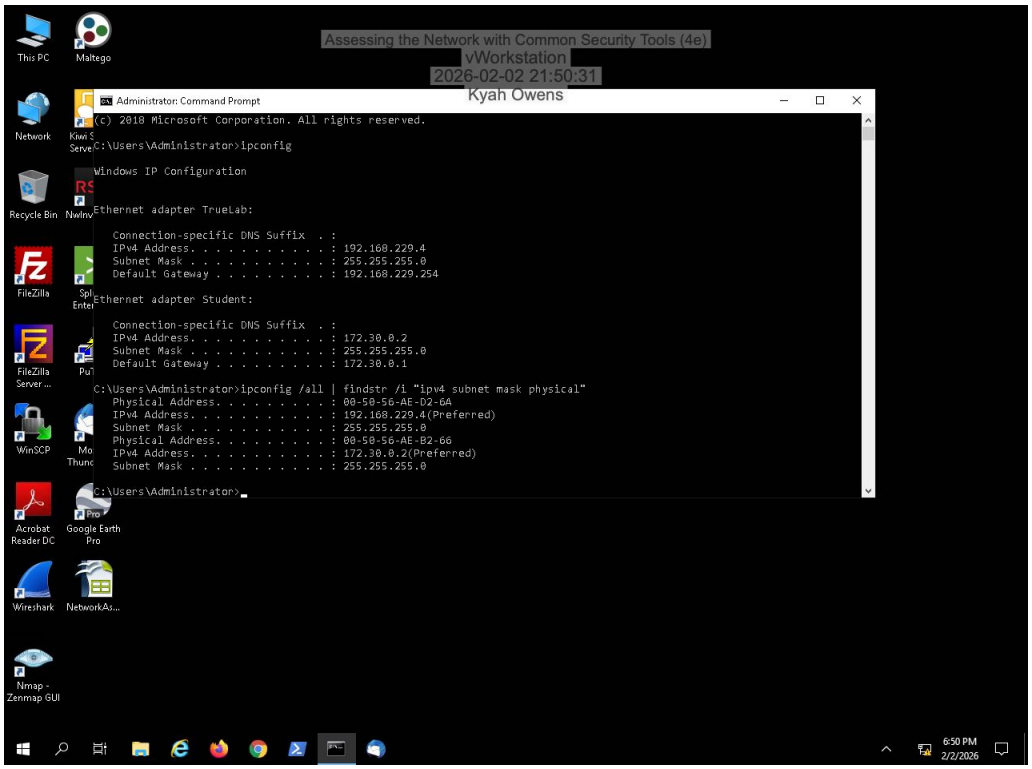
Time on Task:	Progress:
2 hours, 46 minutes	100%

Report Generated: Wednesday, April 15, 2026 at 10:05 AM

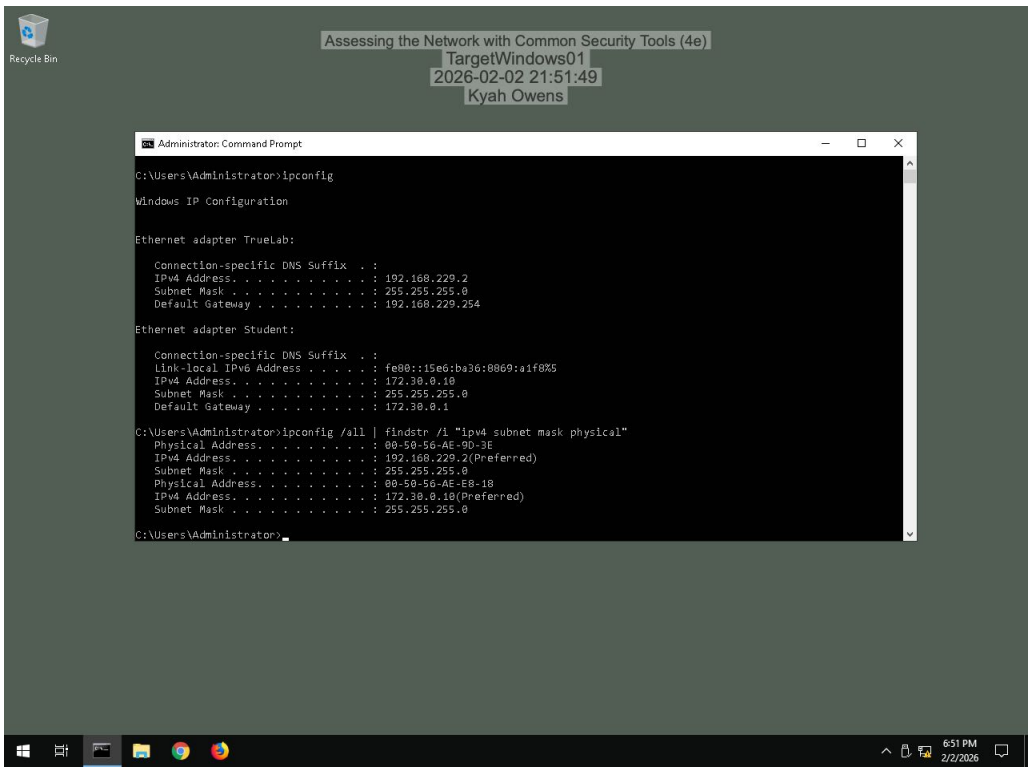
Hands-On Demonstration

Part 1: Exploring the LAN with Basic Network Utilities

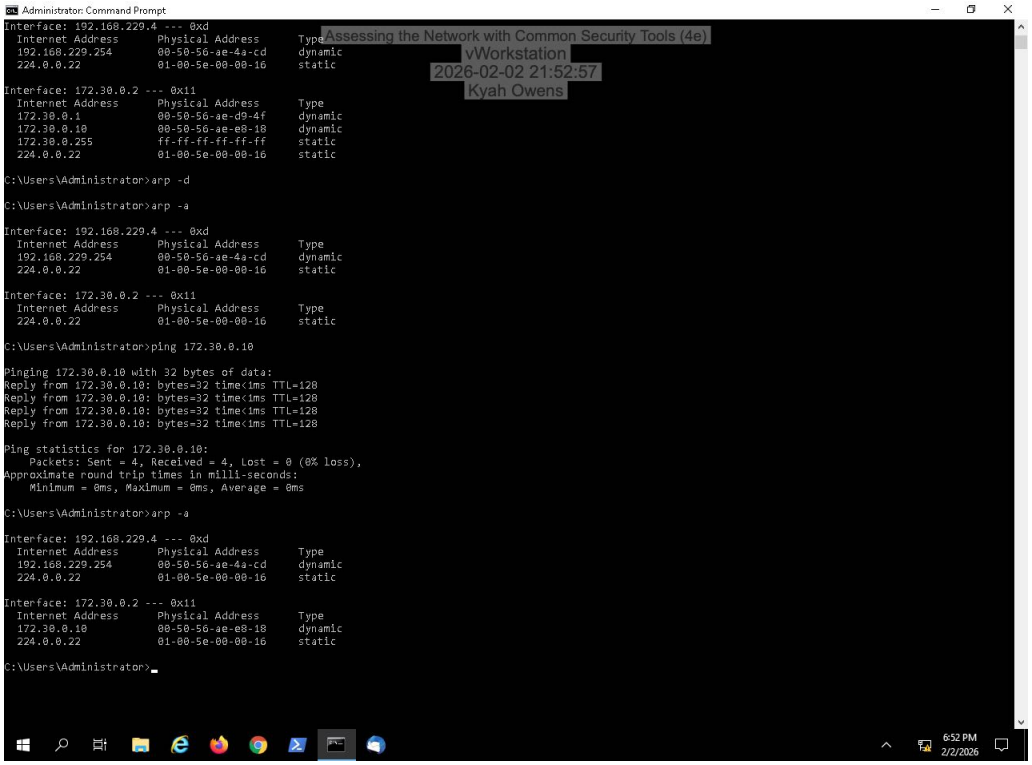
4. **Make a screen capture** showing the **ipconfig results for the Student adapter on vWorkstation**.



7. **Make a screen capture** showing the **ipconfig results for the Student adapter on TargetWindows01**.

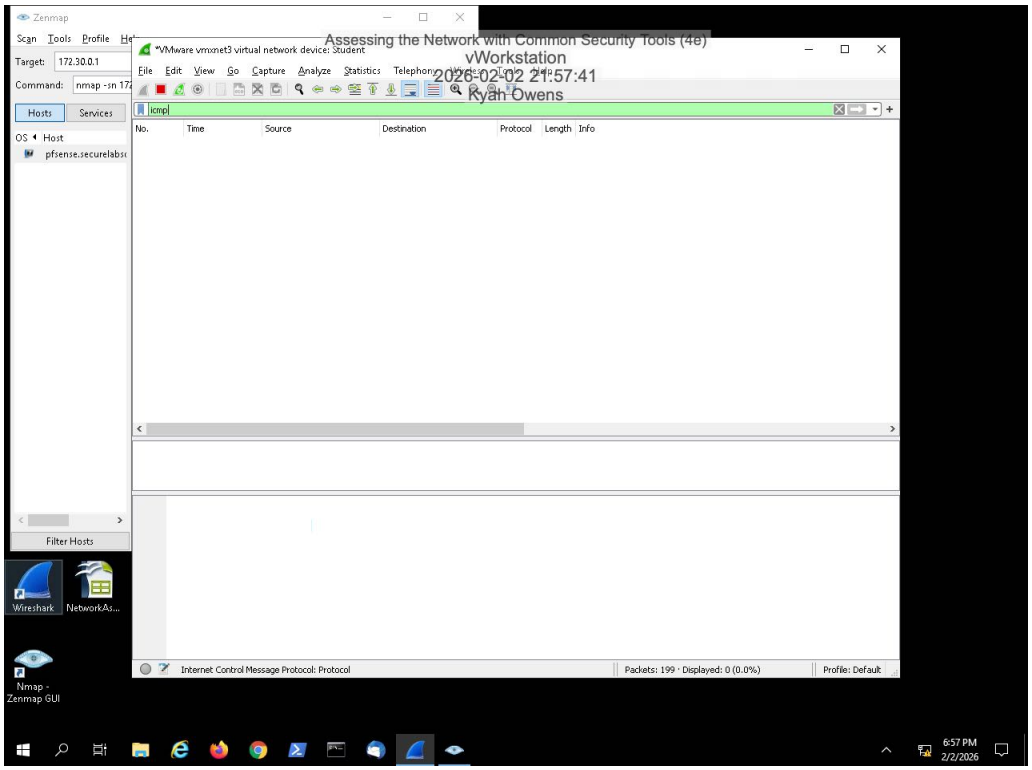


14. **Make a screen capture** showing the **updated ARP cache on vWorkstation**.

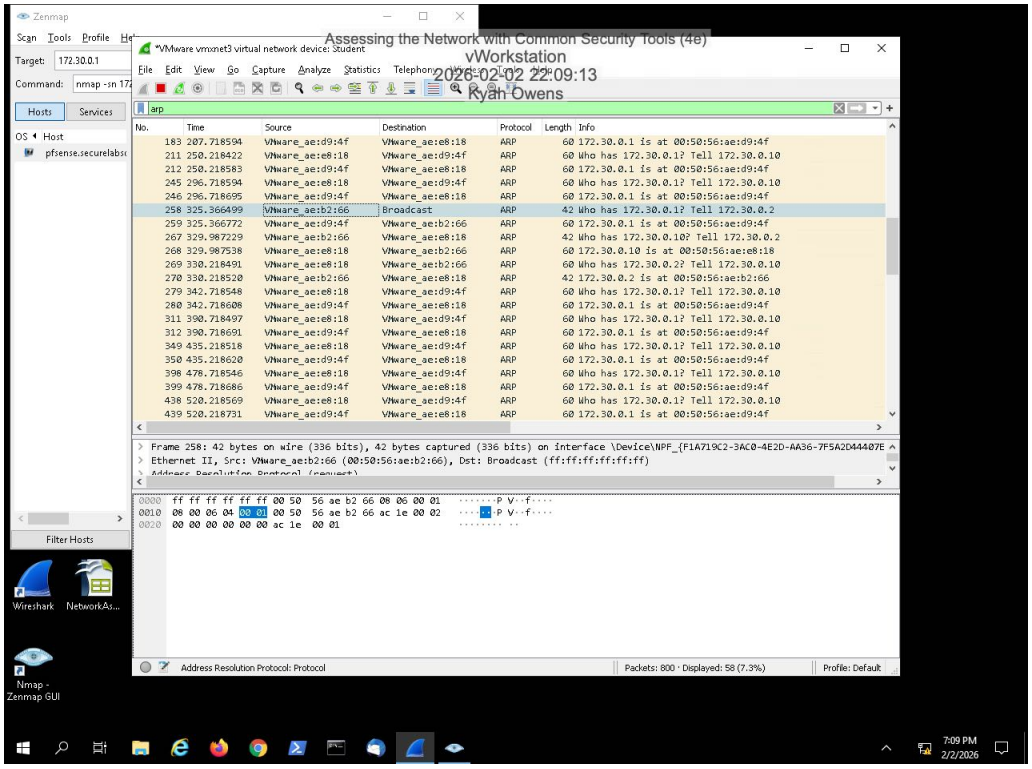


Part 2: Advanced LAN Analysis with Wireshark and Nmap

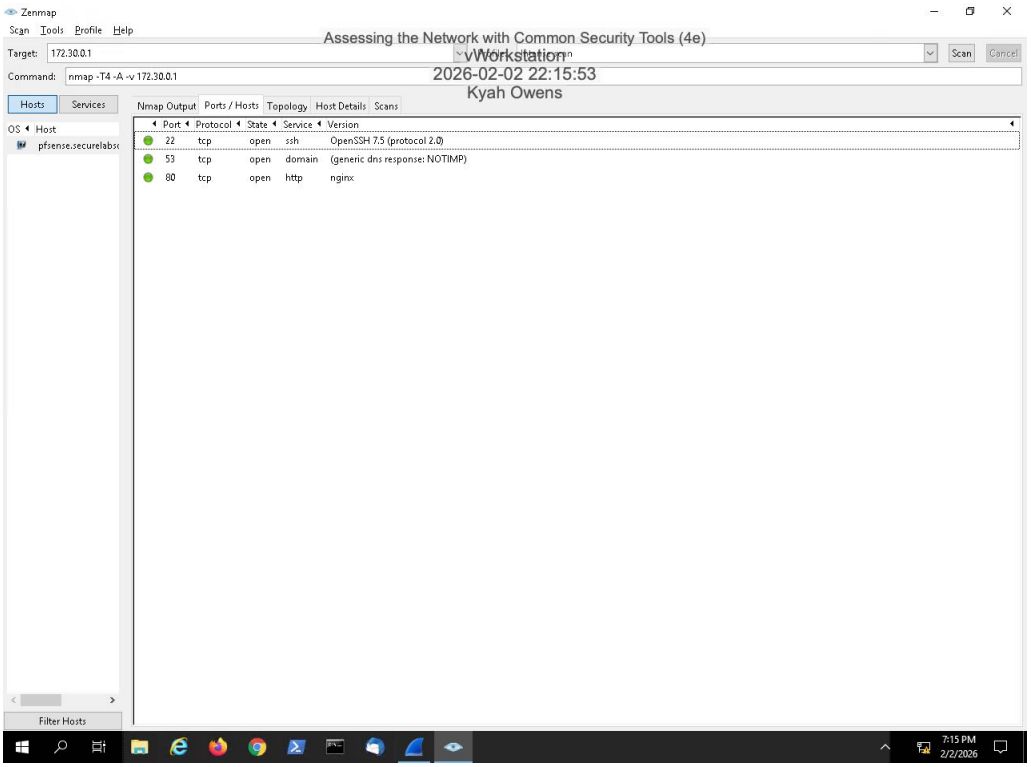
9. **Make a screen capture** showing the **ICMP filtered results in Wireshark**.



12. **Make a screen capture** showing the **ARP filtered results in Wireshark**.



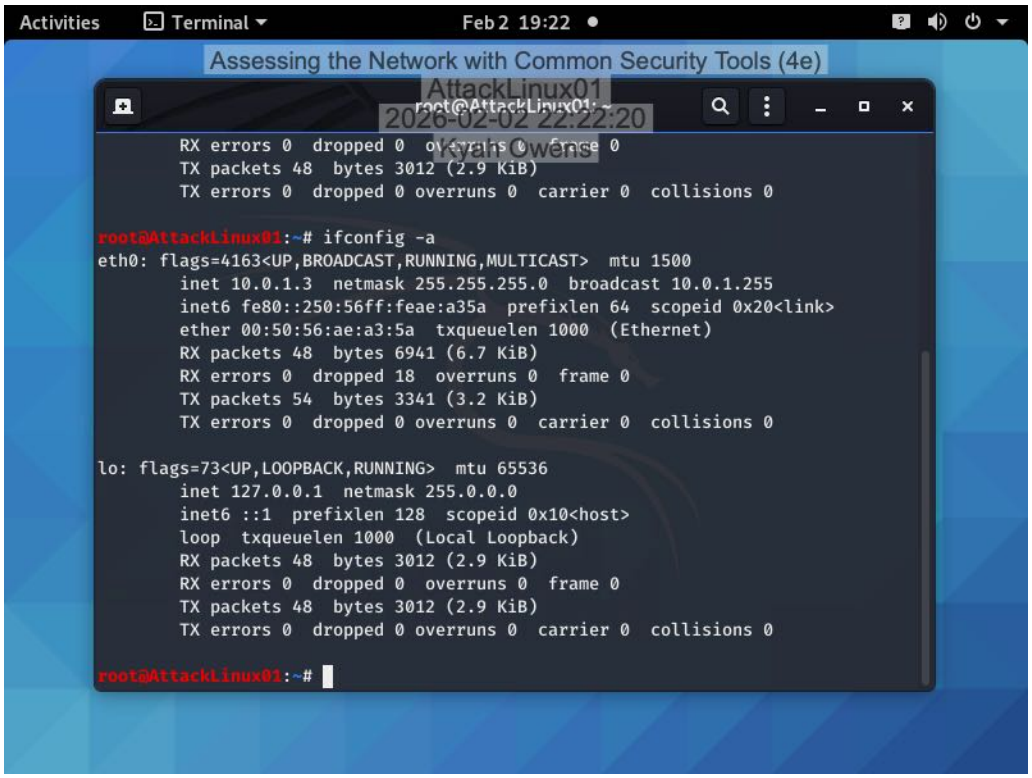
28. **Make a screen capture** showing the **contents of the Ports/Hosts tab**.



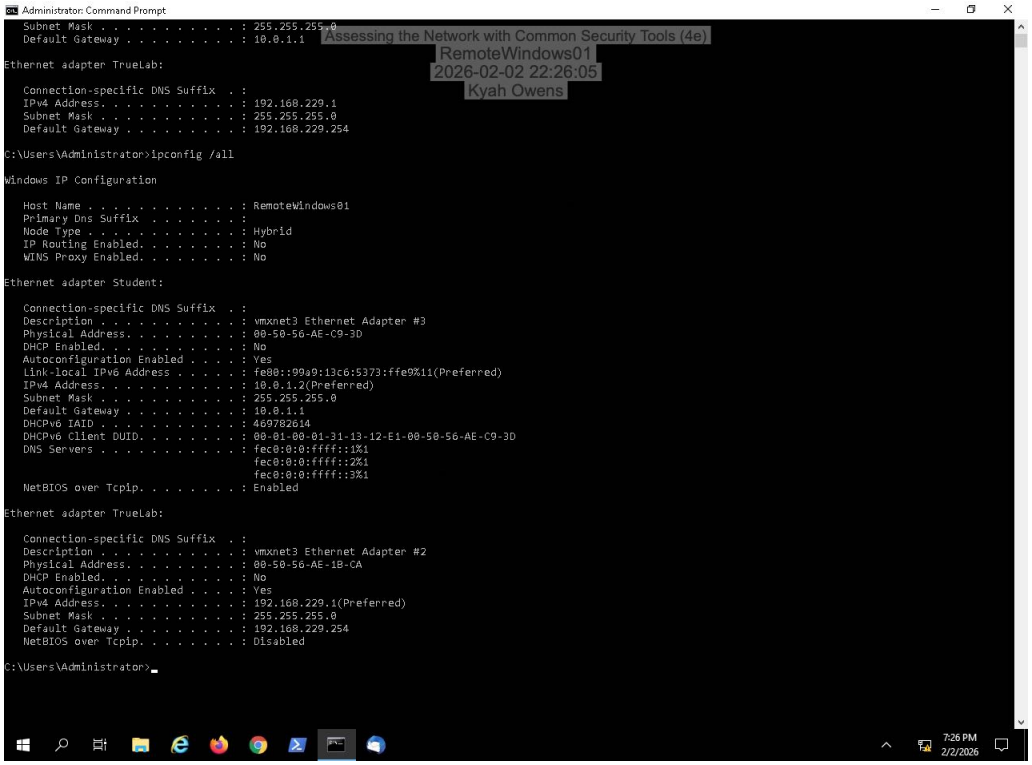
Applied Learning

Part 1: Exploring the WAN

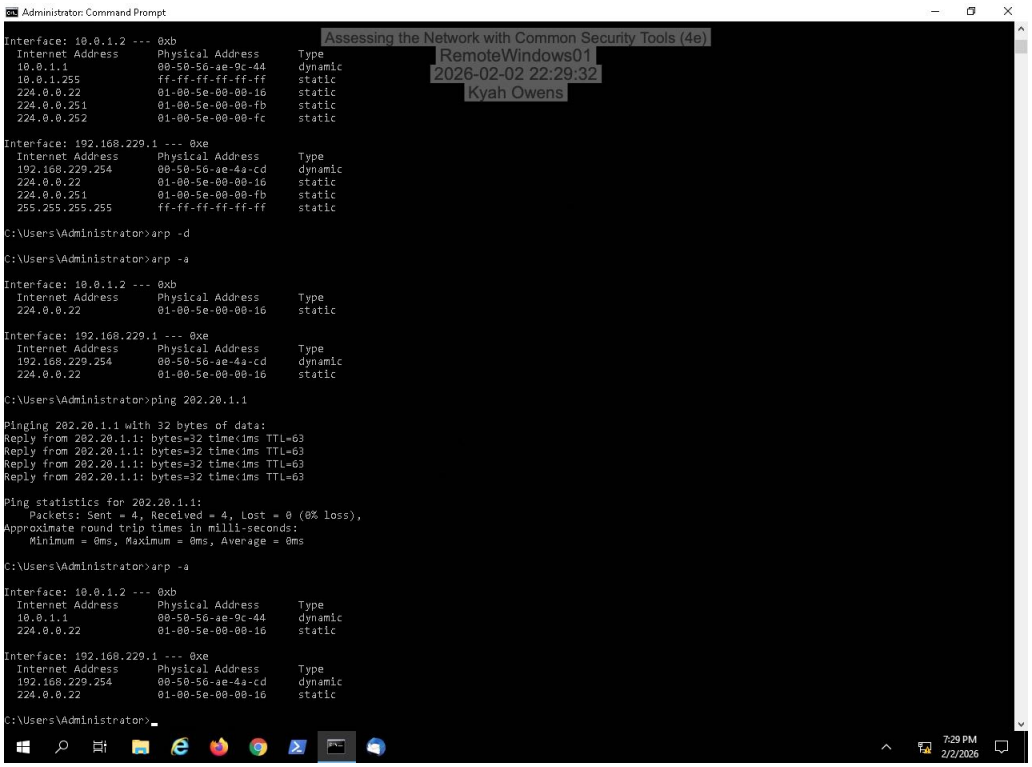
6. **Make a screen capture** showing the **ifconfig results on AttackLinux01**.



12. **Make a screen capture** showing the **ipconfig results on RemoteWindows01**.

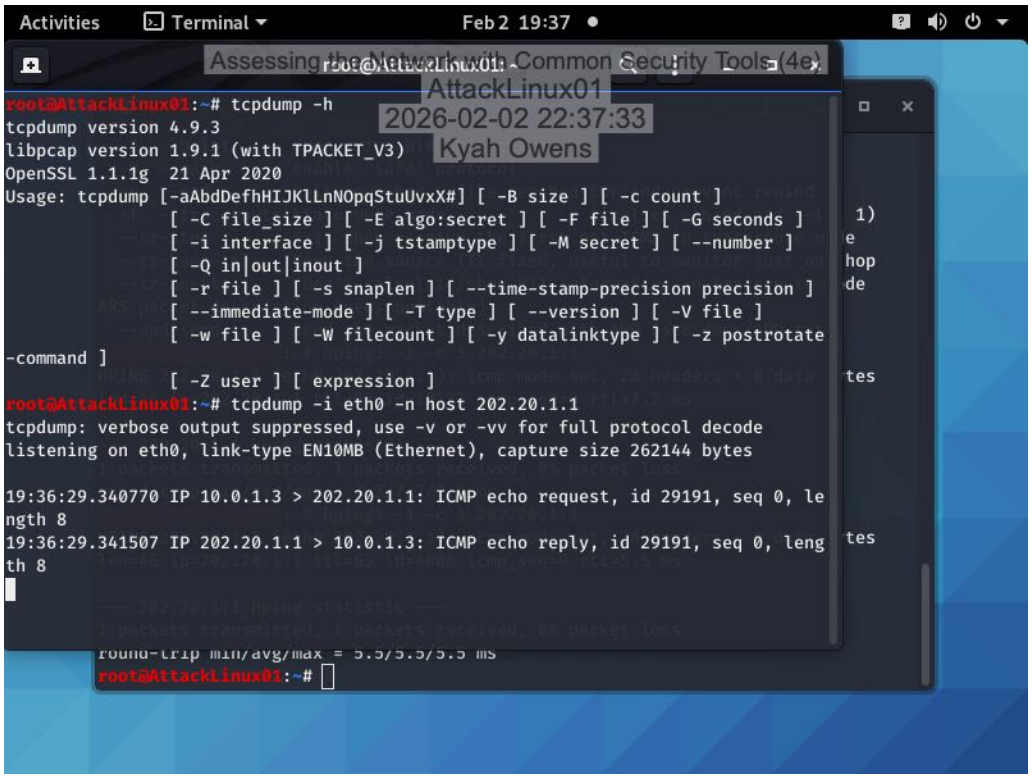


18. **Make a screen capture** showing the **updated ARP cache on RemoteWindows01**.

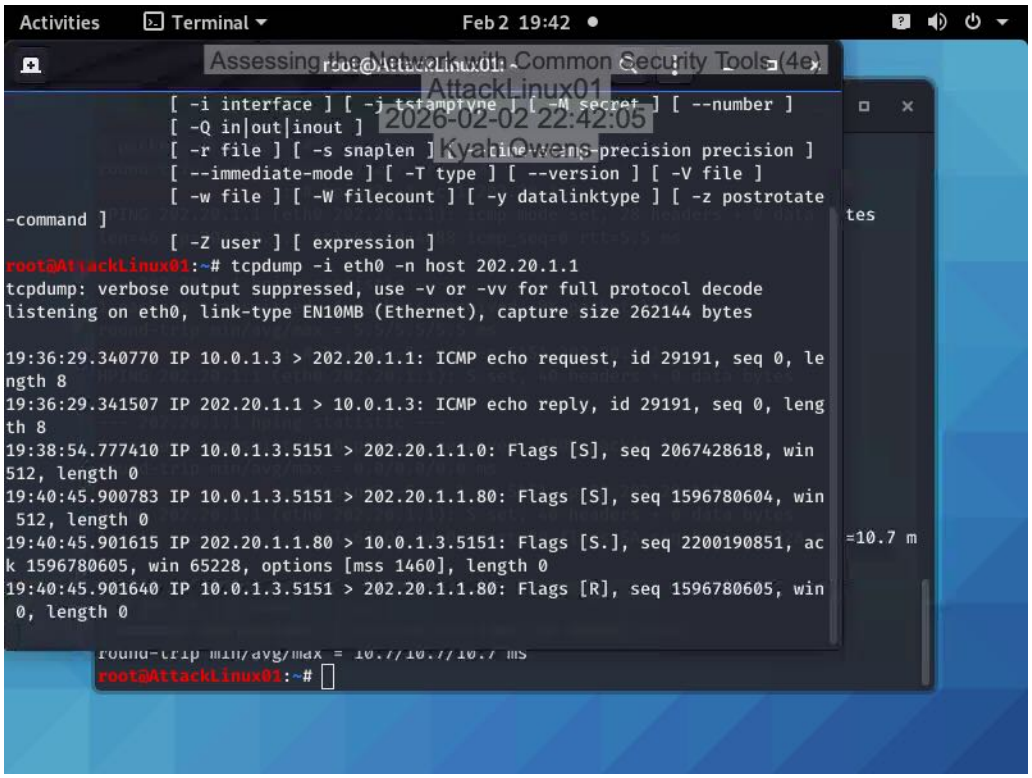


Part 2: Advanced WAN Analysis

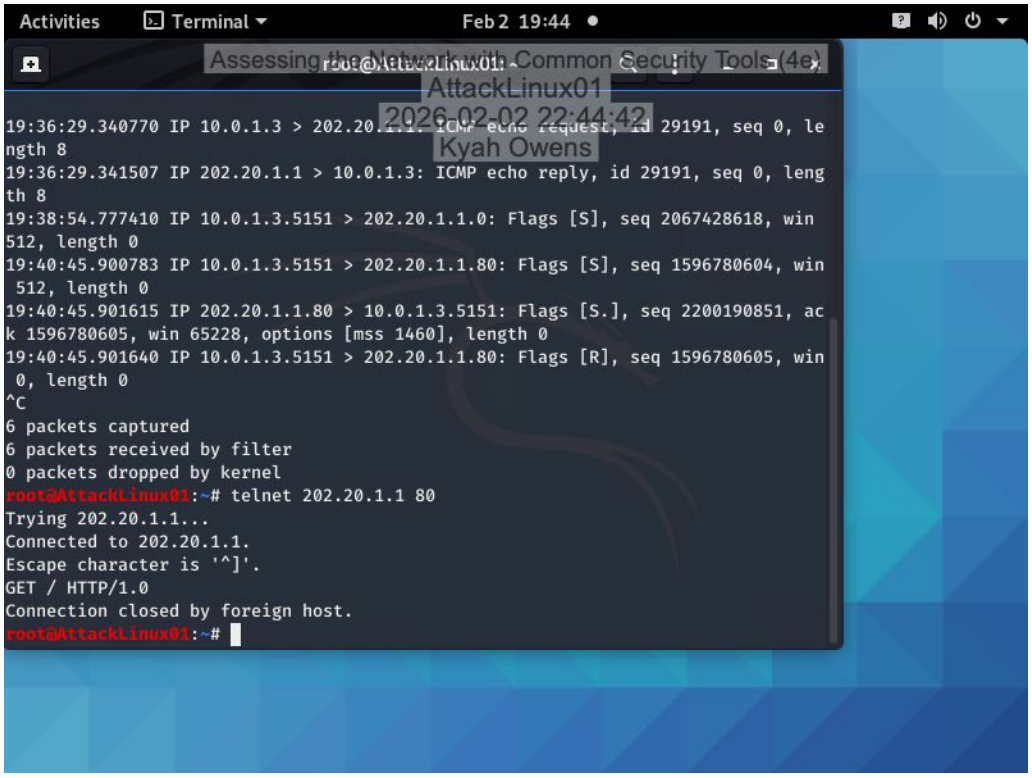
9. **Make a screen capture** showing **tcpdump echo back the captured packets**.



12. **Make a screen capture** showing the **attempted three-way handshake in tcpdump**.



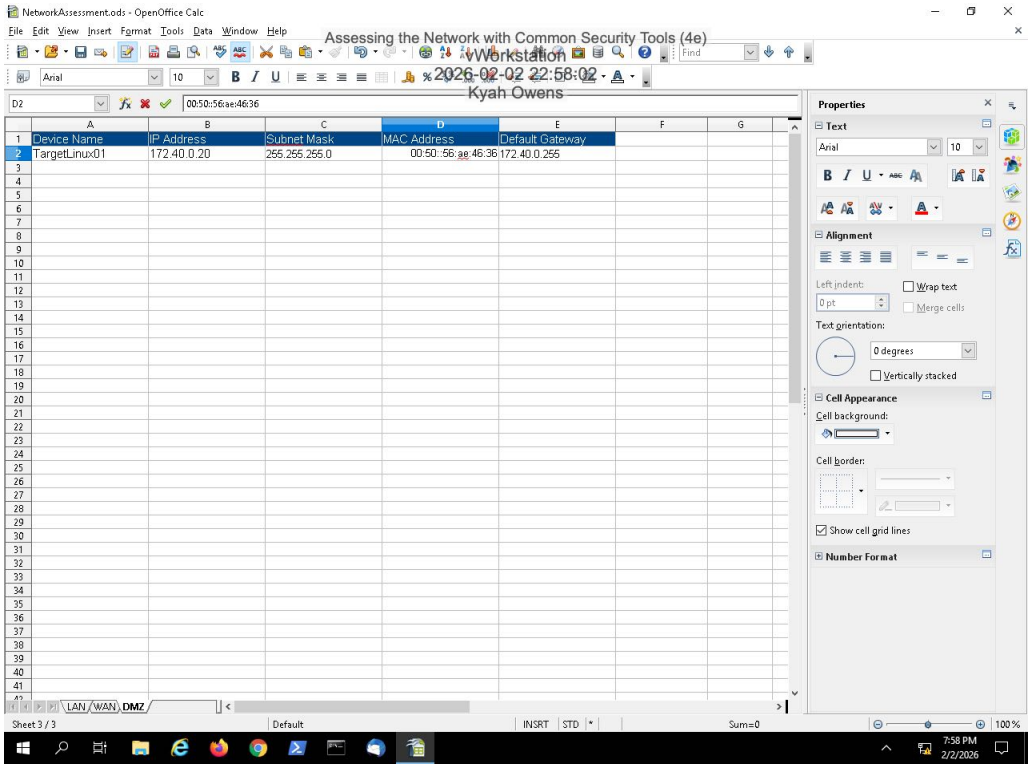
17. **Make a screen capture** showing the **results of the GET command**.



Challenge and Analysis

Part 1

2. **Make a screen capture** showingthe **completed DMZ tab of the NetworkAssessment spreadsheet**.



5. **Briefly summarize** your findings in a technical memo to the CISSM (Chief Information Systems Security Manager).

A network traffic analysis confirmed the pfSense firewall's ability to respond to ICMP traffic and it allows outbound DNS communications. No ARP traffic was seen at the firewall, since ARP communications are confined to the local network segment. An Nmap regular scan revealed TCP ports 22 (SSH) and 80 (HTTP) were open and no additional services were found to be accessible from outside of the network. The above findings demonstrate that the network has a controlled perimeter with very few external services visible.